



SYSTEM PROPOSAL

Lab Equipment Borrowing and Tracking System

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INTRODUCTION

PROJECT CONTENT

The Lab Equipment Borrowing and Tracking System aims to efficiently manage and monitor laboratory equipment through RFID technology. It automates the borrowing and returning process using RFID-tagged equipment and RFID-enabled IDs, reducing manual errors and saving time for students and staff. This system ensures accurate records, enhances accountability, and helps prevent the loss or misplacement of laboratory assets.

PROBLEM STATEMENT

The current process of borrowing and tracking laboratory equipment in schools is often manual and inefficient, leading to issues such as inaccurate record-keeping, unaccounted equipment, and time-consuming transactions. Without an automated system, staff and students face difficulties in ensuring accountability, monitoring equipment availability, and preventing loss or misplacement of laboratory assets. This lack of an organized and reliable tracking method hinders proper utilization of resources and disrupts laboratory operations. Therefore, there is a need for an automated Lab Equipment Borrowing and Tracking System that utilizes RFID technology to streamline the borrowing and returning process, minimize human error, and enhance the overall management of laboratory equipment.)

PROJECT OBJECTIVES

The primary objective of this project is to design and develop a Lab Equipment Borrowing and Tracking System that utilizes Radio Frequency Identification (RFID) technology to automate and enhance the management of laboratory equipment within the school. This system is intended to ensure accurate tracking of borrowed items, promote accountability among users, and reduce manual errors associated with the traditional borrowing process.



Specifically, this study aims to achieve the following objectives:

- To develop a digital system that automates the borrowing and returning of laboratory equipment using RFID-enabled identification cards for both students and faculty members.
- To establish a reliable tracking mechanism capable of recording and updating the real-time status, borrower information, and transaction history of laboratory equipment.
- To minimize manual processes and reduce human errors in recording, monitoring, and verifying laboratory equipment transactions through automated data processing.
- To promote accountability among students and staff in handling laboratory equipment by maintaining accurate, accessible, and tamper-proof borrowing records.
- To provide laboratory administrators with an intuitive management interface that allows efficient monitoring of inventory levels, equipment availability, and usage reports.

REQUIREMENT ANALYSIS

Functional Requirements

These define the specific functions and features that the system must perform to meet its intended objectives.

1. User Authentication and Access Control

- The system shall allow users (students and staff) to log in using their unique RFID-enabled ID cards.
- The system shall provide role-based access (e.g., admin, staff, student).

2. RFID-Based Borrowing and Returning

- The system shall detect and record equipment borrowing when a user scans their RFID ID and the equipment's RFID tag.
- The system shall automatically update the equipment status (e.g., Available, Borrowed, Returned).
- The system shall generate a confirmation message or receipt after each successful transaction.

3. Equipment Management

- The system shall allow administrators to register new equipment with RFID tags.
- The system shall enable updating, editing, or removing equipment records.
- The system shall maintain a record of equipment condition and location.

4. Transaction Monitoring and Logging

- The system shall log all borrowing and returning activities with date, time, user, and equipment details.
- The system shall store historical transaction data for tracking and auditing purposes.



5. Inventory Management

- The system shall display current equipment availability and status.

6. Notification and Alerts

- The system shall send alerts for overdue returns or missing items.
- The system shall notify the administrator of any discrepancies or unregistered equipment detected by the RFID scanner.

7. Reporting Module

- The system shall generate daily, weekly, or monthly reports summarizing borrowing activities, overdue items, and equipment utilization.

Non-Functional Requirements

These describe the quality attributes, performance standards, and constraints of the system.

1. Performance

- The system shall process RFID scans and record transactions in real time (within 2 seconds).

2. Usability

- The system shall provide a user-friendly interface that is easy to navigate for students, staff, and administrators.
- Minimal user training shall be required for operation.

3. Reliability and Availability

- The system shall be operational at least 99% of the time during school hours.
- Data shall be regularly backed up to prevent loss in case of system failure.

4. Security

- The system shall encrypt user credentials and sensitive data.
- Only authorized users shall access administrative functions and reports.

5. Scalability

- The system shall support the addition of more RFID readers and equipment without major system modifications.

6. Maintainability

- The system shall allow for easy updates, bug fixes, and database maintenance.

7. Compatibility

- The system shall be compatible with standard RFID readers and student ID cards used by the school.
- The software shall run on commonly used operating systems (e.g., Windows, Linux).



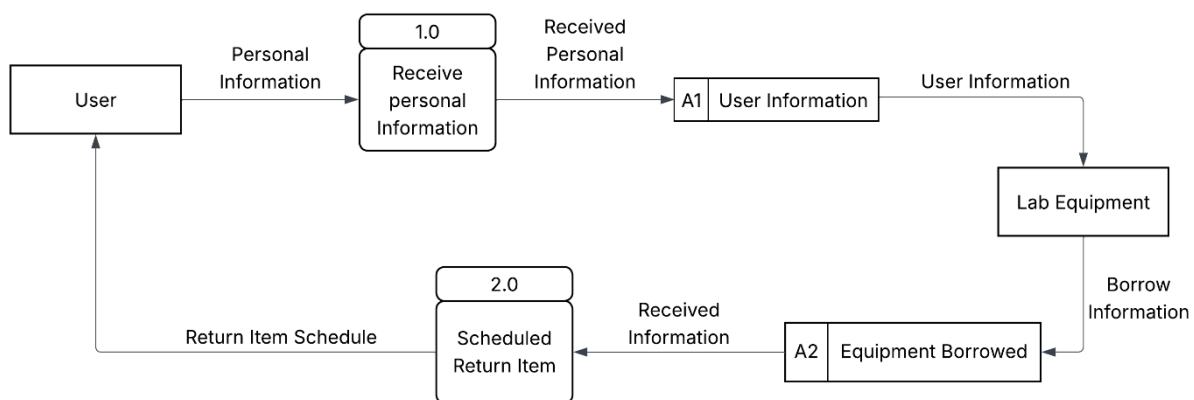
8. Data Integrity

- The system shall ensure that data is accurate, consistent, and synchronized between the RFID reader and the database.

DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) illustrates the overall data movement and processing involved in the Laboratory Equipment Borrowing and Returning System. It depicts how information flows between the user, the system, and the database to facilitate borrowing and returning of laboratory equipment. The DFD is divided into two main processes: the borrowing process and the returning process.

Borrowing Process of Laboratory Equipment



1. Receive Personal Information (Process 1.0)

The system collects and validates the user's personal information. This information is then transmitted to the **User Information (A1)** data store for processing and verification.

2. User Information and Lab Equipment Integration

After verification, the user's information is linked with the corresponding Lab Equipment that the user intends to borrow. The system retrieves the relevant equipment data and associates it with the borrowing user.

3. Equipment Borrowed (A2)

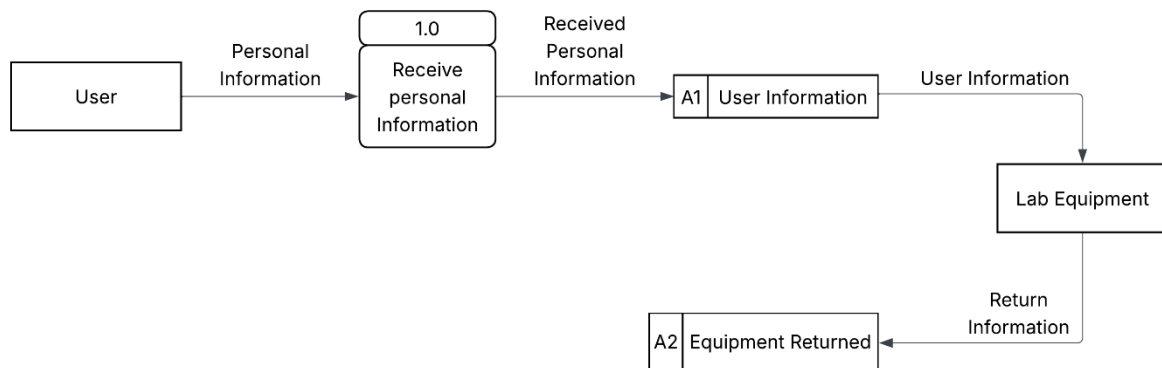
The borrowing details are stored in the **Equipment Borrowed** data store. The system subsequently updates the database to reflect the change in equipment status from "Available" to "Borrowed."

4. Scheduled Return Item (Process 2.0)

The user specifies the intended return date for the borrowed equipment. The system records the **return schedule** and updates the transaction record accordingly, completing the borrowing process.



Returning Process of Laboratory Equipment



1. Receive Personal Information (Process 1.0)

The system receives and verifies the user's information to identify which items were previously borrowed under that user's account.

2. User Information and Lab Equipment Interaction

The verified user data is matched with the corresponding **Lab Equipment** that is being returned.

3. Equipment Returned (A2)

Once the return transaction is confirmed, the system records the event in the **Equipment Returned** data store. The database is then updated to change the equipment status from "*Borrowed*" to "*Available*."